

$$\text{Intensity} = I \cos^2 \theta = I \cos^2(60.0^\circ) = 0.250 I$$

16

D

Since intensity, I varies proportionately with (amplitude, A)²

Let each unit for displacement be y .

$$\frac{I_{\text{new}}}{I_{\text{old}}} = \left(\frac{A_{\text{new}}}{A_{\text{old}}} \right)^2$$

$$\frac{0.5I_{\text{old}}}{I_{\text{old}}} = \left(\frac{A_{\text{new}}}{4y} \right)^2$$

$$A_{\text{new}} = 2.8y$$

17

C